

4COSC002W Mathematics for Computing

Lecture 10

Elementary Statistics

UNIVERSITY OF
WESTMINSTER

What is Statistical Analysis?

A Tool for Insight: Utilising methods to collect, analyse, and interpret data.

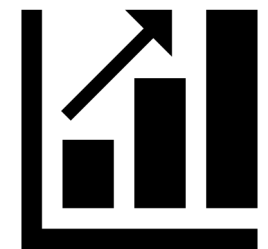
Purpose: To uncover patterns, trends, and insights in data.

Why It Matters in Computing?

- **Informs Decisions:** Guides data-driven decisions in software development and system design.
- **Enhances Performance:** Crucial for evaluating and improving technology performance and user experience.

Software Tools: R, Python, MATLAB, Excel.

“Good decisions come from good data analysis.”



Sample Dataset: Bugs Found in Software Projects

Project	Bugs Found
A	12
B	7
C	9
D	15
E	5
F	8
G	7
E	11

*check the Excel file on Blackboard

Basic Statistical Analysis for the Sample Dataset

Mean	9.250
Median	8.500
Mode	7
Range	10
Standard Deviation	3.240
Variance	10
Interquartile Range (IQR)	4.750

Mean (Arithmetic Average)

The mean is the average value of a dataset, representing the central point of the data.

General Formula: $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$, where x_i are the values in the dataset and n is the number of values.

Excel Formula: =AVERAGE(B2:B9)

Application: In computing, the mean is used to calculate average performance metrics, such as average response time or average number of bugs in software projects.

Example: Sample Dataset Calculations

Mean (Arithmetic Average)

Sum of all values: $12 + 7 + 9 + 15 + 7 + 5 + 8 + 11 = 74$

Number of values: 8

$$\text{Mean: } \frac{74}{8} = 9.25$$

Result: Mean = 9.25

Median

The median is the middle value in a sorted list of numbers, effectively separating the dataset into two halves.

General Approach: If the number of observations is odd, the median is the middle number. If even, it's the average of the two middle numbers.

Excel Formula: =MEDIAN(B2:B9)

Application: The median is helpful in computing for analysing skewed data, such as median load times or median server response times.

Example: Sample Dataset Calculations

Median

Sorted Data: [5, 7, 7, 8, 9, 11, 12, 15]

Number of values: 8 (even)

Median is average of 4th and 5th values: $\frac{8+9}{2} = 8.5$

Result: Median = 8.5

Mode

The mode is the most frequently occurring value in the dataset. It can be used for both numerical and categorical data.

General Approach: Identify the value that appears most frequently.

Excel Formula: =MODE.SNGL(B2:B9)

Application: In computing, the mode can help identify the most common error types, the most used software feature, or the most common number of daily user logins.

Example: Sample Dataset Calculations

Mode

Frequency of each value: 5 (once), 7 (twice), 8 (once), 9 (once), 11 (once), 12 (once), 15 (once)

Most frequent value: 7

Result: Mode = 7

Measures of Variability

Measures of variability describe how spread out or dispersed the data values are in a dataset. They complement measures of central tendency (like the mean or median) by showing the degree to which data points differ from each other and the centre.

- Range
- Interquartile Range (IQR)
- Variance
- Standard Deviation
- Mean Absolute Deviation (MAD)

Why Are Measures of Variability Important?

- Understand *data consistency* (e.g., test scores, production quality).
- Compare *different datasets* (e.g., one with higher variability vs. one with lower variability).
- Identify datasets with potential *outliers*.

An **outlier** is a data point that is significantly *different or far away from the majority of the data* in a dataset. Outliers are extreme values that can occur due to variability in the data, errors in measurement, or unusual circumstances.

Range

The range is the difference between the highest and lowest values in a dataset.

General Approach: Range = Maximum value - Minimum value.

Excel Formula: =MAX(B2:B8) - MIN(B2:B8)

Application: Range is used in computing to understand the spread of data, such as the range of response times or the range in the number of daily active users.

Example: Max Value is 15, Min Value is 5

Range = $15 - 5 = 10$

Standard Deviation

Standard deviation quantifies the amount of variation or dispersion of a set of data values from the mean.

General Formula (for a sample): $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$.

Excel Formula: =STDEV.S(B2:B9) for a sample

Application: Standard deviation is used in computing to measure variability in system performance, such as the consistency of response times or variability in daily website traffic.

What Standard Deviation (SD) Tells Us

The standard deviation measures **how spread out** the data is around the mean:

Small SD:

- Data is *close to the mean*.
- The dataset is *consistent* and has *low variability*
- Predictable, clustered data (e.g., similar test scores)

Large SD:

- Data is *spread out* and varies significantly from the mean.
- The dataset has *high variability*.
- Diverse, scattered data (e.g., a wide range of incomes).

Zero SD:

- All data points are *the same* (no spread).



Example: Sample Dataset Calculations

Standard Deviation

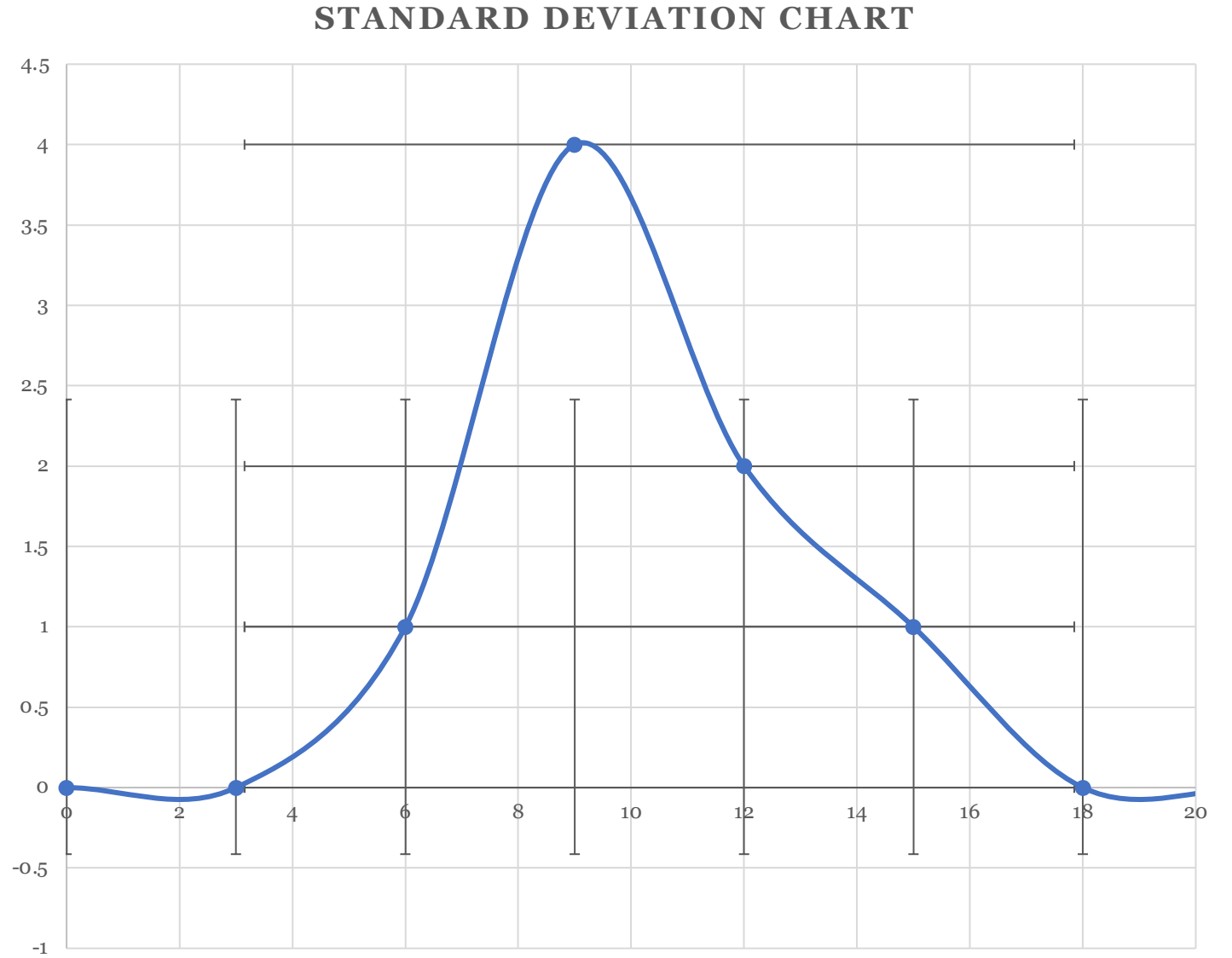
$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

$$s = \sqrt{\frac{(12 - 9.25)^2 + (7 - 9.25)^2 + (9 - 9.25)^2 + (15 - 9.25)^2 + (5 - 9.25)^2 + (8 - 9.25)^2 + (7 - 9.25)^2 + (11 - 9.25)^2}{8 - 1}}$$

$$s = \sqrt{\frac{7.5625 + 5.0625 + 0.0625 + 33.0625 + 18.0625 + 1.5625 + 5.0625 + 3.0625}{7}} = 3.24$$

Result: Standard Deviation = 3.24

Example: Sample Dataset Plot



Variance

Variance looks at the squared differences between each data point and the mean.

General Formula (for a sample): $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$.

Excel Formula: =VAR.S(B2:B9) for a sample

Application: Variance is important in computing for identifying the consistency of data, like the consistency in the number of bugs found across different projects.

Why Use Both SD and Variance?

- Variance is useful for statistical and mathematical purposes (e.g., in formulas or when combining data from multiple sources).
- Standard deviation is better for interpretation because it is in the *same units as the data*.

Example: Sample Dataset Calculations

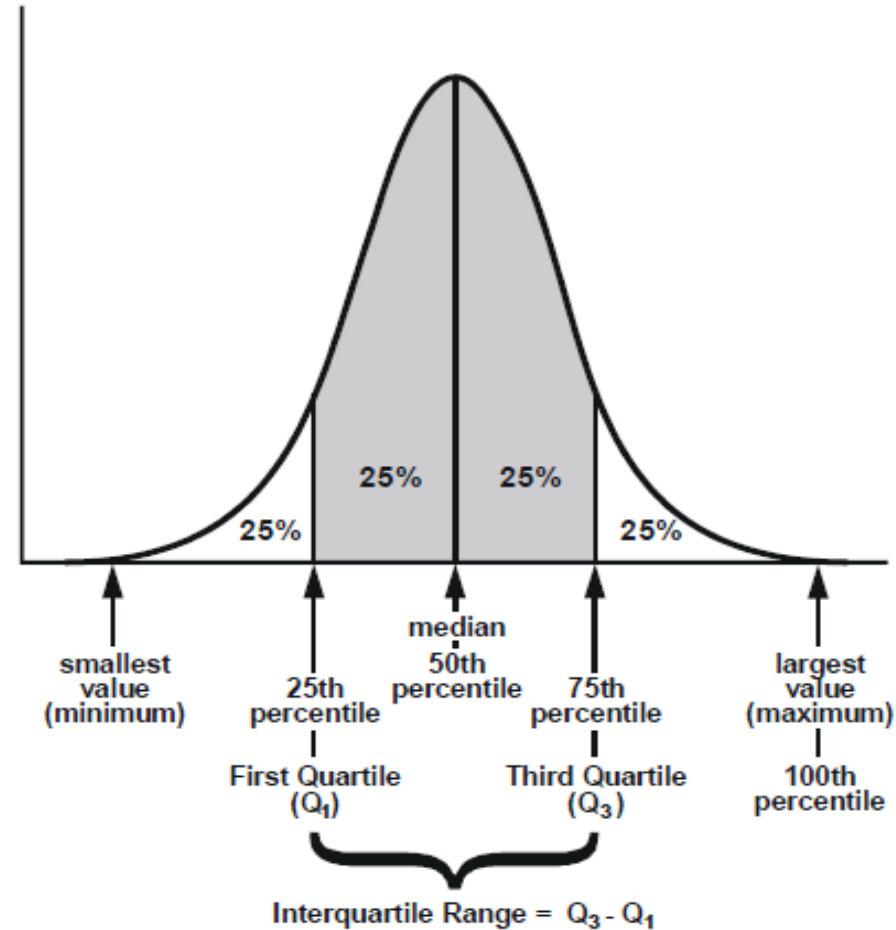
Variance

Variance is the *square of the standard deviation*.

$$\text{Variance} = (\text{Standard Deviation})^2 = 3.24^2$$

Result: Variance = 10.0

IQR vs Normal Distribution



What Are Quartiles?

Quartiles divide a dataset into **four equal parts**, each containing 25% of the data. They help summarise the distribution and identify the spread and centre of the data.

Types of Quartiles:

1. First Quartile (Q1):

- The value below which 25% of the data falls.
- It marks the *25th percentile*.

2. Second Quartile (Q2):

1. The value below which 50% of the data falls (the *median*).
2. It marks the *50th percentile*.

3. Third Quartile (Q3):

1. The value below which 75% of the data falls.
2. It marks the *75th percentile*.

Why Are Quartiles Important?

- Measure Spread: They show how the data is distributed.
- Identify Outliers: Help find extreme values using the Interquartile Range (IQR).
- Compare Datasets: Useful for spotting differences in spread and central tendency.

Interquartile Range (IQR)

IQR is the range between the first quartile (25th percentile) and the third quartile (75th percentile). It shows the middle 50% of the data. Provides reliable variability without being skewed by outliers.

Percentiles are measures indicating the value below which a given percentage of observations fall.

General Approach:

$$\text{IQR} = Q_3 - Q_1,$$

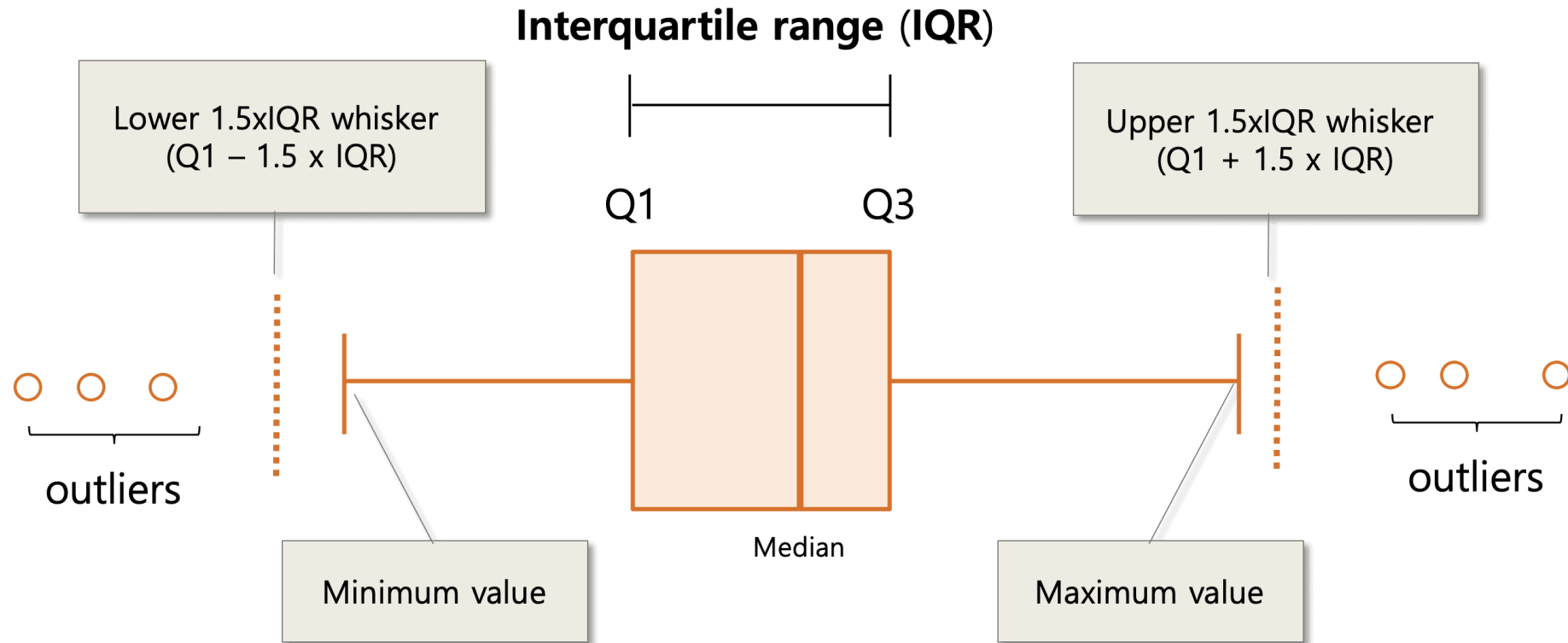
where Q_3 and Q_1 are the third and first quartiles, respectively.

Excel Formula: `=QUARTILE.EXC(B2:B9, 3) - QUARTILE.EXC(B2:B9, 1)`

Why is IQR Needed?

- Measures Core Spread: Focuses on the middle 50% of data, ignoring extremes.
- Resistant to Outliers: Provides reliable variability unaffected by outliers.
- Outlier Detection: Helps find unusual values for data cleaning.
- Works for Skewed Data: Ideal for non-normal distributions.

IQR Boxplot



Step-by-Step Guide to Calculate the Interquartile Range (IQR)

1. Sort the Data:

- Arrange the data in ascending order.

2. Find the Quartile Positions:

- For $Q1$ (25th percentile): Position = $\frac{(n+1)}{4}$
- For $Q3$ (75th percentile): Position = $\frac{3(n+1)}{4}$

3. Determine Quartile Values:

- If the quartile position is an integer, use the value at that position.
- If the position is not an integer, use interpolation: $Q = \text{Value at } k + f \times (\text{Value at } k + 1 - \text{Value at } k)$, where:
 - k is the integer part of the position.
 - f is the fractional part of the position.

4. Calculate IQR: $\text{IQR} = Q3 - Q1$

Example: Sample Dataset Calculations

IQR Calculation

Data (Bugs Found): 12, 7, 9, 15, 5, 8, 7, 11

Step 1: Sort the Data

Sorted Data: [5, 7, 7, 8, 9, 11, 12, 15]

Step 2: Find the Quartile Positions

Total number of values (n) = 8.

Position for $Q1$: $\frac{8+1}{4} = 2.25$

Position for $Q3$: $\frac{3(8+1)}{4} = 6.75$

Example: Sample Dataset Calculations

Step 3: Calculate $Q1$ and $Q3$

For **Q1** (Position = 2.25):

$k = 2, f = 0.25$.

Value at $k = 7$, Value at $k + 1 = 7$ (look at the sorted list).

Interpolation: $Q1 = 7 + 0.25 \times (7 - 7) = 7$

For **Q3** (Position = 6.75):

$k = 6, f = 0.75$.

Value at $k = 11$, Value at $k + 1 = 12$.

Interpolation: $Q3 = 11 + 0.75 \times (12 - 11) = 11 + 0.75 = 11.75$

Step 4: Calculate IQR: $IQR = Q3 - Q1 = 11.75 - 7 = 4.75$

Example: Sample Dataset IQR Boxplot Plot

